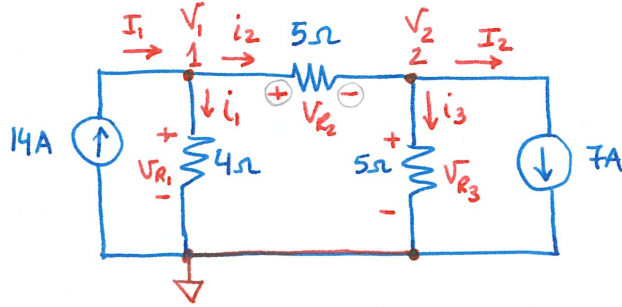


EXERCISE 8 (Nodal)



Obtain node voltages and power dissipated in the 4Ω resistor

① Identify nodes and choose ground as reference node

② Assign voltages V_1 and V_2 to nodes 1 and 2

③ Apply KCL to nodes 1 and 2 (label circuit first!)

Node 1. $I_1 = i_1 + i_2$

Node 2. $i_2 = i_3 + I_2$

$$\Rightarrow 14A = \frac{V_{R1}}{4} + \frac{V_{R2}}{5} \Rightarrow 14 = \frac{V_1}{4} + \frac{V_1 - V_2}{5}$$

$$\Rightarrow \frac{V_{R2}}{5} = \frac{V_{R3}}{5} + 7 \Rightarrow \frac{V_1 - V_2}{5} = \frac{V_2}{5} + 7$$

Express currents in terms of node voltages

$$\frac{V_1}{4} + \frac{V_1}{5} - \frac{V_2}{5} = 14; \quad V_1 \left(\frac{1}{4} + \frac{1}{5} \right) - \frac{V_2}{5} = 14; \quad \frac{9}{20} V_1 - \frac{1}{5} V_2 = 14 \quad (\times 2)$$

$$\frac{V_1}{5} - \frac{V_2}{5} - \frac{V_2}{5} = 7; \quad \frac{V_1}{5} - \frac{2}{5} V_2 = 7; \quad \frac{1}{5} V_1 - \frac{2}{5} V_2 = 7$$

$$\left(\frac{18}{20} - \frac{1}{5} \right) V_1 + 0 = 21 \Rightarrow \frac{18-4}{20} V_1 = 21; \quad \frac{14}{20} V_1 = 21$$

$$V_1 = \frac{20 \times 21}{14} = \underline{\underline{30V}}$$

Substitute V_1 in one of the equations:

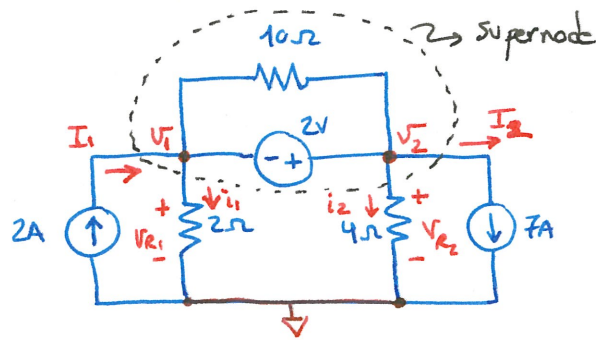
$$\frac{30}{5} - \frac{2}{5} V_2 = 7 \Rightarrow -\frac{2}{5} V_2 = 7 - 6; \quad V_2 = \underline{\underline{-\frac{5}{2}V}}$$

* Power dissipated in 4Ω resistor

$$P = \frac{V^2}{R} = \frac{30^2}{4} = \underline{\underline{225W}}$$

EXERCISE 9 (Nodal)

Find the node voltages.



- Write supernode equation (KVL in middle mesh):

$$V_2 - V_1 = 2V \quad (-V_1 - 2V + V_2 = 0)$$

- Apply KCL in supernode:

$$I_1 = i_1 + i_2 + I_2$$

$$2A = \frac{V_{R1}}{R_1} + \frac{V_{R2}}{R_2} + 7A \Rightarrow 2 = \frac{V_1}{2} + \frac{V_2}{4} + 7$$

$$-V_1 + V_2 = 2$$

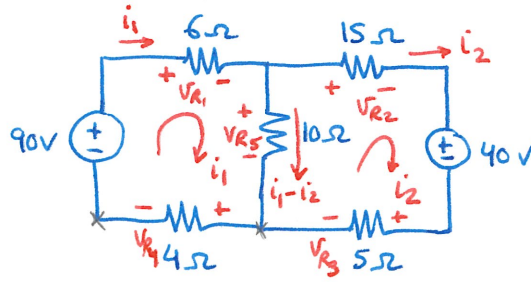
$$\frac{V_1}{2} + \frac{V_2}{4} = -5 \quad (+ \times 2)$$

$$0 + (1 + \frac{1}{2})V_2 = -8 \quad ; \quad \frac{3}{2}V_2 = -8 \Rightarrow V_2 = \frac{-16}{3} = \underline{\underline{-5.333V}}$$

Substitute V_1 in first equation

$$-V_1 - \frac{16}{3} = 2 \Rightarrow V_1 = \frac{-16}{3} - 2 = \underline{\underline{-7.333V}}$$

EXERCISE 10 (Mesh)



Calculate the mesh currents, i_1 and i_2 , and the power in the 40V voltage source.

- ① Identify meshes and assign mesh currents
- ② Apply KVL to each of the n meshes (label circuit first!)

$$-90 + V_{R1} + V_{R5} + V_{R4} = 0 \Rightarrow -90 + 6i_1 + 10(i_1 - i_2) + 4i_1 = 0$$

$$-V_{R5} + V_{R2} + 40 + V_{R3} = 0 \Rightarrow -10(i_1 - i_2) + 15i_2 + 40 + 5i_2 = 0$$

↑
Express voltages in
resistors in terms of mesh currents

Rearrange:

$$\left. \begin{array}{l} -90 + 6i_1 + 10i_1 - 10i_2 + 4i_1 = 0 \\ -10i_1 + 10i_2 + 15i_2 + 40 + 5i_2 = 0 \end{array} \right\} \begin{array}{l} 20i_1 - 10i_2 = 90 \\ -10i_1 + 30i_2 = -40 \quad (\times 2) \end{array}$$

$$0 + 50i_2 = 10$$

$$\hookrightarrow i_2 = \frac{10}{50} = \underline{\underline{0.2 \text{ A}}}$$

Substitute i_2 in one of the equations:

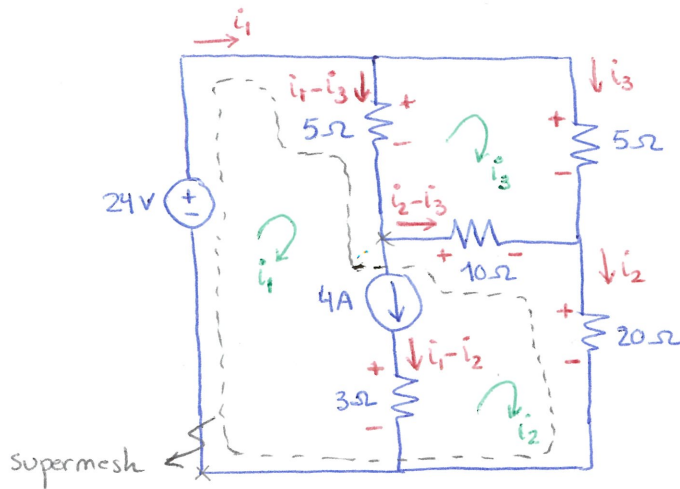
$$20i_1 - 10 \times 0.2 = 90; \quad 20i_1 - 2 = 90 \quad i_1 = \frac{92}{20} = \underline{\underline{4.6 \text{ A}}}$$

* Power in 40V voltage source

$$P = +40\text{V} \times i_2 = +8\text{W} \text{ (dissipated)}$$

EXERCISE 11 (Mesh)

Find i_1 , i_2 and i_3 .



- Write supermesh equation

$$i_1 - i_2 = 4 \text{ A}$$

- KVL in:

- supermesh: $-24 \text{ V} + 5(i_1 - i_3) + 10(i_2 - i_3) + 20i_2 = 0$

- mesh 3 (top left): $-5(i_1 - i_3) + 5i_3 - 10(i_2 - i_3) = 0$

Simplify:

$$i_1 - i_2 = 4$$

$$5i_1 + (10 + 20)i_2 + (-5 - 10)i_3 = 24$$

$$-5i_1 - 10i_2 + (5 + 5 + 10)i_3 = 0$$

$$\left. \begin{array}{l} i_1 - i_2 = 4 \quad \text{①} \\ 5i_1 + 30i_2 - 15i_3 = 24 \quad \text{②} \\ -5i_1 - 10i_2 + 20i_3 = 0 \quad \text{③} \end{array} \right\}$$

① $i_1 = 4 + i_2$

Substituting in ② and ③:

$$\left. \begin{array}{l} 5(4 + i_2) + 30i_2 - 15i_3 = 24 \\ -5(4 + i_2) - 10i_2 + 20i_3 = 0 \end{array} \right\}$$

$$\left. \begin{array}{l} 35i_2 - 15i_3 = 4 \quad \text{④} \\ -15i_2 + 20i_3 = 20 \quad \text{⑤} \end{array} \right\}$$

$$\begin{array}{r} 700i_2 - 300i_3 = 80 \\ -225i_2 + 300i_3 = 300 \end{array}$$

$$475i_2 = 380 \Rightarrow \underline{i_2 = 0.8 \text{ A}}$$

Substituting i_2 in ④

$$35 \times 0.8 - 15i_3 = 4; \quad 28 - 15i_3 = 4; \quad -15i_3 = -24 \Rightarrow \underline{i_3 = \frac{-24}{-15} = 1.6 \text{ A}}$$

Finally, $\underline{i_1 = 4 + i_2 = 4.8 \text{ A}}$